perplexity

i need to show like what results was obained from their research and what are we tyring to do

Absolutely! Here's a clear academic comparison you can present, showing **what results were obtained in "Self-Refine" and "Tree of Thoughts"** (their main findings) and **what your proposed project aims to achieve beyond or differently:**

Base Paper Comparison Table

Aspect	Self-Refine (Madaan et al., NeurIPS 2023)	Tree of Thoughts (Yao et al., ICML 2023)	Your Project (Project Phoenix)
Goal	Enable LLMs to iteratively improve outputs by generating, critiquing, and refining without human feedback	Develop agentic structured reasoning where an LLM explores multiple solutions as a tree, self-selecting	Build a production-ready agentic platform capable of full runtime self-correction, adaptive learning, and autonomous multi-agent collaboration (no human)
Key Results	Self-feedback loops improve solution quality, outperforming "single-shot", vanilla LLM, and few-shot baselines	Outperforms regular CoT and single-pass LLMs in tasks like arithmetic, planning, code, logical puzzles	Will extend self-correction to complex, real-world tasks involving modules for error detection, RL, adaptation, and coordination at scale, not shown in the original papers
Correction Mechanism	"Generate → Critique → Revise → Repeat", all agent-triggered, NO human in loop	Branching, self-evaluating exploration with state backtracks, pruning, autonomous candidate evaluation	Unified loop for pattern/anomaly detection, rule-based/ML corrections, automated rollback and recovery with persistent memory
Memory & Adaptation	Basic context tracking, local self-refinement	Expands "memory" by exploring/retaining multiple intermediate pathways	Implements persistent short-term & long-term memory, meta-learning, continuous learning, and experience-based adaptation
Multi-Agent Support	Single agent-centric	Implied via tree search but single LLM agent instantiation	Decentralized, protocol-driven multi-agent collaboration & consensus for large-scale orchestration

Aspect	Self-Refine (Madaan et al., NeurIPS 2023)	Tree of Thoughts (Yao et al., ICML 2023)	Your Project (Project Phoenix)
Autonomy Level	Fully autonomous, feedback loop-driven	Fully autonomous, search tree agent-driven	100% autonomous: no “human feedback” or annotation; all modules (learning, correction, coordination) are agentic
Evaluation	Standard NLP benchmarks (code generation, QA, arithmetic, commonsense)	Arithmetic, logical puzzles, planning, creative tasks (benchmark suite)	Will be evaluated on real-world system metrics: error detection rate, recovery speed, task throughput, system robustness, no human oversight
What You Add/How Yours Differs	Domain-agnostic, extensible to specialized fields (code, reasoning, scientific docs); larger-scale; full production deployment; advanced monitoring, metricization, operational tools		
Paper Venue	NeurIPS 2023, arXiv:2303.17651	ICML 2023, arXiv:2305.10601	N/A - Proposed. Builds and extends upon both above.

Sample Text to Present:

> - **“In Self-Refine, Madaan et al. show that large language models can self-correct by generating feedback and using it to refine their own answers, resulting in significantly more accurate outputs on code, reasoning, and general QA benchmarks compared to models without self-correction. This was demonstrated across multiple tasks with quantitative improvements.”**
> - **“Tree of Thoughts (Yao et al.) demonstrates that by letting the agent branch, evaluate, and select among alternate solution paths—without human-in-the-loop—models outperform traditional ‘chain-of-thought’ approaches on arithmetic, planning, and creative reasoning benchmarks.”**
> - **“Our project (‘Phoenix’) aims to operationalize and generalize these ideas in a complete, production-grade platform: with persistent memory, robust meta-learning and adaptation, advanced error/recovery systems, and truly autonomous, protocol-based coordination between multiple specialized agents. Unlike the base papers which focus on core concepts/demonstrations, we will engineer, test, and validate an end-to-end agentic runtime framework suitable for industrial/developer deployment—without any manual feedback or intervention.”**

Summary

Use this table and summary to show your professor exactly how your work is **grounded in, but extends beyond the results in those base papers**—especially with respect to end-to-end autonomy, scalability, and production-readiness. If you want a 1-page written justification as a report/slide, just say so!

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1. Project-Phoenix-5.pdf
2. Project-Phoenix-4.pdf
3. Project-Phoenix-3.pdf
4. Project-Phoenix-2.pdf
5. Project-Phoenix-1.pdf